

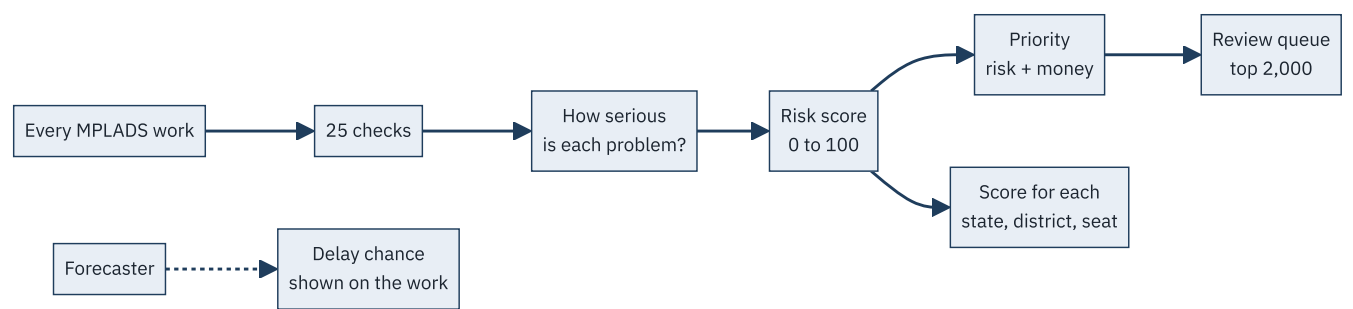
How MPLADS Review Scores Risk – in Simple Words

Aaryan Jain · 24 September 2026 · Every formula from the full formulas document, explained in plain words

The system does two separate jobs. A **rulebook** checks every work and raises a flag when something is wrong or unusual; only the rulebook can flag a work. A **forecaster** looks at a newly recommended work and estimates how likely it is to be delayed; it gives advice but never flags anything.

Each formula below is shown exactly as in the full document, then explained: **what it means**, **what the symbols are**, and a small **example**.

The big picture



Part 1: the rulebook

Step 1 – Run the checks

Every work goes through 25 checks. Each check asks one question, for example:

Check	The question it asks
Delay in Sanction	Did this work take much longer to be approved than similar works?
Prolonged Delay	Has it crossed a fixed time limit (for example, a year without approval)?
Excess Expenditure	Was more money paid than was sanctioned?
Unusual Cost	Does it cost far more than similar works in the same state?
Duplicate Work	Was the same work recommended twice?
Agency Concentration	Is one agency getting most of a district's money?
Completion Evidence	Was it marked complete and paid, with no proof attached?

When a check finds a problem, it creates a **finding**: a note saying what is wrong, with the evidence.

Step 2 – How serious is each problem? (the severity score)

Formula 1 – the three levels

low : $0 \leq x < 0.3333$

medium : $0.3333 \leq x < 0.6667$

high : $0.6667 \leq x \leq 1$

- **What it means:** every finding gets a seriousness score x between 0 and 1, like a speedometer. The first third is "low", the middle third "medium", the top third "high".
- **Symbols:** x = the seriousness score.

Formula 2 – how x is worked out (a straight-line ramp)

$$x = x_a + \frac{v - v_a}{v_b - v_a} (x_b - x_a) \quad \text{for } v_a \leq v \leq v_b$$

- **What it means:** each check measures something (days, a ratio, a cost). It has a few fixed "marker points". Between two markers, x rises in a straight line. Below the first marker x stays at its lowest; above the last it stays at its highest.
- **Symbols:** v = what was measured for this work; v_a and v_b = the two markers it falls between; x_a and x_b = the scores at those markers.
- **Example (fixed time limit of 365 days):** the markers are $1\times$ the limit $\rightarrow \frac{1}{3}$, $1.5\times \rightarrow \frac{2}{3}$, $3\times \rightarrow 1$. A work at $1.2\times$ the limit sits 40% of the way from $1\times$ to $1.5\times$, so $x = \frac{1}{3} + 0.4 \times (\frac{2}{3} - \frac{1}{3}) = 0.47$ (medium).

The markers for each check:

Check	What is measured	Markers (measure $\rightarrow x$)
Peer-comparison delays (6 checks)	days, against similar works in the same state, activity and Lok Sabha term	slower than 90% of peers $\rightarrow 0$, 95% $\rightarrow \frac{1}{3}$, 99% $\rightarrow \frac{2}{3}$, well beyond $\rightarrow 1$
Prolonged Delay (fixed limits: 365 days to sanction, 365 days unsanctioned, 540 days open)	days $\div$ limit	$1\times \rightarrow \frac{1}{3}$, $1.5\times \rightarrow \frac{2}{3}$, $3\times \rightarrow 1$
Excess Expenditure; Sanction Exceeds Recommendation	paid $\div$ sanctioned, or sanctioned $\div$ recommended	$1.01 \rightarrow \frac{2}{3}$, $2.0 \rightarrow 1$
Unusual Cost	the robust z-score (Formula 3)	$5 \rightarrow 0$, $6.5 \rightarrow \frac{1}{3}$, $13 \rightarrow \frac{2}{3}$, $19.5 \rightarrow 1$
Expenditure After Completion	days from completion to last payment	$30 \rightarrow 0$, $180 \rightarrow \frac{1}{3}$, $540 \rightarrow \frac{2}{3}$
Agency Concentration	agency's share of district payments	top 10% of shares $\rightarrow 0$, top 5% $\rightarrow \frac{1}{3}$, top 1% $\rightarrow \frac{2}{3}$, 100% $\rightarrow 1$
Payment and Completion Mismatch	gap between paid and completed amount $\div$ completed	$0.05 \rightarrow 0$, $0.5 \rightarrow \frac{1}{3}$
Trust/Society Limit	total $\div$ ₹1 crore cap	$1\times \rightarrow \frac{1}{3}$, $2\times \rightarrow \frac{2}{3}$
Allocation Limit	overage $\div$ allocation	$0.05 \rightarrow \frac{1}{3}$, $0.5 \rightarrow \frac{2}{3}$
Unspent Balance	$1 -$ share of funds used	$0.4 \rightarrow 0$, $1.0 \rightarrow \frac{1}{3}$

Check	What is measured	Markers (measure → x)
Checks with a fixed level (Expenditure Without Sanction, Duplicate Work, Prohibited Work, Completion Evidence, Record Date, Calamity Consent)	—	the middle of their level: 1/6, 1/2 or 5/6

Formula 3 — how "unusual" a cost is (robust z-score)

$$z = \frac{0.6745 (\log_{10} \text{cost} - \text{median}_{\text{peers}})}{\max(\text{MAD}_{\text{peers}}, 0.05)}$$

- **What it means:** how many "typical spreads" this work's cost sits above the usual cost of similar works. Costs are compared on a log scale, so "twice as expensive" means the same for a ₹1 lakh work and a ₹1 crore work.
- **Symbols:** median = the middle cost of similar works (same state, activity and term); MAD = how spread out those costs usually are. The 0.05 floor stops a group where almost every work has the same price from calling a tiny difference "unusual". 0.6745 is a standard constant that puts z on the familiar statistical scale.
- **Example:** z = 6.5 means the cost is far outside the normal spread; that is where the check starts calling it medium.

When the fixed-limit check and the peer-comparison check flag the same delay on one work, they become one finding with the higher score (11,561 duplicates merged).

Step 3 — Two fairness adjustments to the score

Formula 4 — "looks unusual" can't be high on its own

$$x' = 0.5 + (x - 0.5) \cdot \frac{0.6666 - 0.5}{1 - 0.5} \quad (x > 0.5)$$

- **What it means:** a statistical finding (one that only "looks unusual") is squeezed into the top half of medium. The strongest ones still stay ahead of weaker ones, but none reaches high by itself.
- **Symbols:** x = the score before, x' = the score after. It applies only above 0.5.
- **Example:** x = 0.90 becomes 0.5 + 0.4 × (0.1666 ÷ 0.5) = 0.63 (top of medium).

Formula 5 — agreeing evidence raises it one level

$$x' = \min\left(1, x + \frac{1}{3}\right)$$

- **What it means:** if the same work also has a *different kind* of problem, its strongest statistical finding goes up one level (a third), but never past 1.
- **Example:** 0.63 becomes 0.96 (high).

Officers' reviews can also step in. A type of flag with 50 or more reviews that officers often reject can't be high. A pattern that officers dismiss at least 80% of the time is hidden.

Step 4 — How much each problem counts (strength)

Formula 6 — strength

$$s = \text{curve}(x) \times c, \quad \text{curve} : (0, 0.10) \left(\frac{1}{6}, 0.15\right) \left(\frac{1}{2}, 0.35\right) \left(\frac{5}{6}, 0.60\right) (1, 0.70)$$

- **What it means:** the seriousness score is turned into a weight, then multiplied by how much we trust the check.
- **Symbols:** curve = a straight-line lookup through the five points shown ($x = 0$ gives 0.10, $x = \frac{1}{2}$ gives 0.35, $x = 1$ gives 0.70, and so on); c = trust: **1.0** for a hard rule break (such as paid more than sanctioned), **0.6** for "looks unusual". A paperwork slip is capped at 0.05.
- **Example:** a medium statistical finding at $x = \frac{1}{2}$ gives $s = 0.35 \times 0.6 = \mathbf{0.21}$. A high rule finding at $x = \frac{5}{6}$ gives $s = 0.60 \times 1.0 = \mathbf{0.60}$.

Step 5 – The Risk score of a work (0 to 100)

Formula 7 – Risk

$$\text{Risk} = 100 \left[1 - \prod_f (1 - s_f^{\max}) \right]$$

- **What it means:** Risk is $100 \times$ the chance that at least one problem is real. Problems are grouped into kinds (time, money, rules, concentration, paperwork, data entry). Only the strongest problem of each kind counts, so three delay flags on one stuck work don't pile up, but a delay plus a money problem do.
- **Symbols:** f = each kind of problem; $s_f^{\max}$ = the strongest problem of that kind; $\prod$ = multiply these together.
- **Example:** a delay (0.60) and an unusual cost (0.21): $\text{Risk} = 100 \times [1 - (1 - 0.60)(1 - 0.21)] = 100 \times (1 - 0.316) = \mathbf{68.4}$.

Step 6 – Priority: adding the money

Formula 8 – money factor E and Priority

$$E = \min \left(1, \max \left(0, \frac{\log_{10}(\text{amount}/100,000)}{2} \right) \right)$$

$$\text{Priority} = \text{Risk} \times (0.5 + 0.5E)$$

- **What it means:** E grows from 0 for a ₹1 lakh work to 1 for a ₹1 crore work (and stays at 1 above that). Priority keeps between half and all of Risk, depending on the money, so a risky big work comes before an equally risky small one.
- **Symbols:** amount = sanctioned amount, or recommended amount if not yet sanctioned.
- **Example:** ₹70 lakh gives $E = \log_{10}(70) \div 2 = 0.92$, so $\text{Priority} = 68.4 \times (0.5 + 0.46) = \mathbf{65.8}$. The review queue shows the 2,000 highest-priority works of ₹5 lakh or more in the current Lok Sabha.

Step 7 – Scores for states, districts, constituencies and agencies

A place's score combines **how often** its works are flagged with **how serious** those flags are. Only real problems count: a work whose only issue is missing paperwork is not counted as flagged here.

Formula 9 – the national rate and how much to trust small places

$$p_0 = \frac{\sum_i k_i}{\sum_i n_i} \quad \tau^2 = \frac{\sum_i n_i (k_i/n_i - p_0)^2 - (R-1) p_0 (1 - p_0)}{N - \sum_i n_i^2 / N} \quad m = \frac{p_0 (1 - p_0)}{\tau^2} - 1$$

- **What it means:** p_0 is the share of all works flagged across the country. τ^2 measures how much places *really* differ, after removing the differences you'd expect by chance. m is "how many extra average works to add to each place" before judging it: if places barely differ, m is big (lean on the national rate); if they differ a lot, m is small (trust each place's own record).
- **Symbols:** k_i = flagged works in place i ; n_i = all works in place i ; R = number of places; N = all works in total.
- **Result:** $m = 80$ for states, 7.4 for districts, 8.0 for constituencies (always kept between 5 and 1,000).

Formula 10 – the fair flagged rate

$$\text{shrunk_rate}_i = \frac{k_i + p_0 m}{n_i + m}$$

- **What it means:** the place's flagged rate, after adding m "average" works to it. A place with very few works gets pulled toward the national rate; a place with thousands barely moves. It's like judging a new batter on more than one innings.
- **Example:** national rate 18%, $m = 8$, a seat with 3 works of which 2 flagged: $(2 + 0.18 \times 8) \div (3 + 8) = 31\%$, not a misleading 67%.

Formula 11 – money-weighted Risk and the final place score

$$\text{weighted_risk}_i = \frac{\sum_{w \in i} \text{amount}_w \cdot \text{Risk}_w}{\sum_{w \in i} \text{amount}_w}$$

$$\text{risk_score}_i = \text{shrunk_rate}_i \times \text{weighted_risk}_i$$

- **What it means:** the average Risk of the place's flagged works, where bigger works count for more. The final score = how often $\times$ how serious.
- **Symbols:** w = each flagged work in place i . The "breach rate" shown on dashboards is simply $k_i \div n_i$.

Part 2: the forecaster (delay prediction)

On the day an MP recommends a work, the forecaster estimates the **chance it will be seriously delayed**. It learned from 90,838 past 17th Lok Sabha works recommended from 1 April 2023, whose outcome is known.

Formula 12 – what counts as "seriously delayed"

$$G_s = P_{90}(\text{sanction date} - \text{recommendation date}) = 382 \text{ days}$$

$$G_e = P_{90}(\text{completion date} - \text{sanction date}) = 702 \text{ days}$$

- **What it means:** the slowest 10% of past works took more than 382 days to be approved, or more than 702 days to be finished after approval. Those two numbers are the delay lines.
- **Symbols:** P_{90} = the 90th percentile (90% of works were faster).

Formula 13 – how each past work was labelled for learning

$$y = \begin{cases} 1 & \text{sanction lag} > G_s \vee (\text{unsanctioned} \wedge t - \text{rec. date} > G_s) \\ & \vee \text{execution lag} > G_e \vee (\text{open} \wedge t - \text{sanction date} > G_e) \\ 0 & \text{sanctioned and completed, sanction lag} \leq G_s, \text{ execution lag} \leq G_e \\ \text{excluded} & \text{otherwise (too young to judge; 2,062 works)} \end{cases}$$

- **What it means:** a past work counts as **delayed (1)** if it crossed either delay line, including works still waiting that have already been waiting longer than the line. It counts as **on time (0)** if it was approved and finished within both lines. A work too new to judge is left out rather than guessed.
- **Symbols:** t = the data snapshot date (10 September 2026); $\vee$ = "or"; $\wedge$ = "and".

The 12 things it looks at, all known on the day of recommendation, most useful first:

#	Input	How it is calculated
1	How common the district is	works in this district ÷ all past works
2	How common the state is	works in this state ÷ all past works
3	Amount	$\log_{10}(\text{recommended amount})$
4	How common the activity is	works of this activity ÷ all past works
5	Letter batch size	works sent in the same recommendation letter
6	Description length	characters in the work description
7	District office workload	works the district authority received in the 90 days before
8	Cost vs similar works	$\log_{10}(\text{amount}) - \text{the middle } \log_{10}(\text{amount}) \text{ of similar works}$
9	MP workload	works the same MP recommended in the 30 days before
10	Month	month of recommendation
11	Year-end window	1 if January, February or March, else 0
12	Quarter	calendar quarter of recommendation

Formula 14 – the predicted chance

$$p = g \left(\sigma \left(\sum_{k=1}^{600} f_k(\mathbf{x}) \right) \right)$$

- **What it means:** 600 small decision trees each look at the work's 12 inputs and add a little evidence for or against a delay. The total is turned into a 0–1 chance, then adjusted so it matches reality: a p of 0.64 means about 64 out of 100 such works really were delayed.
- **Symbols:** $\mathbf{x}$ = the work's 12 inputs; f_k = the k -th tree; σ = the standard "squash into 0 to 1" function; g = the reality-matching adjustment (calibration).

Formula 15 – the tiers

High : $p \geq P_{90}(p_{\text{past}}) = 0.641$ Medium : $p \geq P_{60}(p_{\text{past}}) = 0.391$ Low : otherwise

- **What it means:** High = the riskiest 10% of past works; Medium = the next 30%; Low = the remaining 60%.

Tier	Who gets it	How often they were really delayed
High	the riskiest 10% of works	about 7 in 10
Medium	the next 30%	about 5 in 10
Low	the other 60%	about 2 in 10

Formula 16 – the reasons shown with a prediction

$$\text{show input } j \iff x < \text{sub} > j < / \text{sub} > \geq P_{90}(j) \vee x < \text{sub} > j < / \text{sub} > \leq P_{10}(j) \quad (\text{u})$$

- **What it means:** an input is listed as a reason when this work is in the top 10% or bottom 10% of all past works for it, for example "Recommended amount is in the top 10% of all works".

How good it is. It was trained on older works and tested on newer ones it had never seen, which is how it will be used. It scored **0.73** on a scale where 0.5 is a coin toss and 1.0 is perfect. On average about 36% of works are delayed; about 70% of its High-tier works were.

Part 3: the "odd one out" score

Formula 17 – anomaly score

$$\text{anomaly}_w = \frac{\#\{v \in \text{group}(w) : \text{score}(v) > \text{score}(w)\}}{|\text{group}(w)|}$$

- **What it means:** the share of similar works that look *more normal* than this one. Works are compared only within the same group: same stage (just recommended, in progress, or finished) and similar cost, 12 groups in all.
- **Symbols:** w = this work; v = every other work in its group; score = how normal a work looks to an Isolation Forest (a standard "spot the outlier" method); # = "how many".
- **Example:** 0.95 means more unusual than 95% of its peers. This score only reorders works tied on priority in the queue; it never adds or removes a flag.

In one line

Checks find problems → each problem gets a seriousness score (Formulas 1–5) → problems become strengths and combine into a 0–100 Risk (6–7) → money turns Risk into Priority (8) → places are scored fairly (9–11). Separately, the forecaster predicts delays (12–16), and the odd-one-out score breaks ties (17).